ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 – 2022)

TARGET : JEE (M + A) : 2022

ENTHUSIAST SRG TEST SERIES

Test Type: ONLINE | Test Date : 12 - 01 - 2022 | Pattern: JEE advance

Time : 3 Hours | TEST NO - 11 (PAPER 1) | Maximum Marks : 180

Instructions

READ THE INSTRUCTIONS CAREFULLY

QUESTION PAPER FORMAT AND MARKING SCHEME

1. The question paper has **three parts** : **PHYSICS**, **CHEMISTRY** and **MATHEMATICS**. Each part has **THREE SECTIONS**.
2. Carefully read the instructions given at the beginning of each section.
3. **Section I(i):** This section contains **SIX** questions. The answer to each question is a **ONLY ONE CORRECT OPTION.**

   Marking scheme : +3 for correct answer, 0 if not attempted and –1 in all other cases.
4. **Section I(ii):** This section contains **SIX** questions. The Answer to each question is a **ONE OR MORE THAN ONE CORRECT OPTIONS**.

Marking scheme :	+4	for correct answer, 0 if not attempted and –2 in all other cases.
Partial Marks :	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks :	+2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks :	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.

5. **Section II:** This section contains **SIX** questions. The answer to each question is a **SINGLE OR THREE DIGIT INTEGER**.

   Marking scheme : +3 correct answer, 0 if not attempted and 0 in all other cases.

ALLEN SOUTH INDIA CENTERS

ALLEN BENGALURU CAMPUSES

JAYANAGAR | KORAMANGALA | MARATHAHALLI | BANASWADI | HEBBAL | HSR LAYOUT
BANNERGHATTA | BASAVESHWARA NAGAR | SARJAPURA | JALAHALLI | INDIRANAGAR
WHITEFIELD

Tel : 080-46704000

ALLEN MYSURU CENTER

SARASWATHIPURAM CAMPUS
B M. Arcade, No-2923, 1st Main, 5th Cross, Saraswathipuram, Opp. SVC Bank, Mysuru-570009, Karnataka
Tel. 0821-4526818, 9945588588

ALLEN MANGALURU CENTER : (TALLENTPRO)
Vikaas Pre-University College Vikas Group of institutions, Airport Road, Mary Hill Mangaluru, Karnataka-575008
Tel.: 9900090058

ALLEN CHENNAI CAMPUSES

ASHOK NAGAR CAMPUS (H.O.)
No. 93 & 94, Next to Kasi Theatre, Jafferkhanpettai, Ashok Nagar, Chennai-600083, Tamil Nadu.
Tel.: 044-40261000

VELACHERY CAMPUS
Plot No.13, New survey No - 347/2A, Vijaya Nagar, 1st Cross, Main Road, Velachery, Chennai - 600042, Tamil Nadu
Tel: 9116687303, 9116687304

ANNA NAGAR-EAST CAMPUS
New No : 80, Old no : 419, Kilpauk Garden Road, Anna Nagar East, Chennai–600010, Tamil Nadu
Tel.: 9116687305, 9116687306

ADYAR CAMPUS
Above Nalli Silks,City Towers, 7, 3rd Cross St, Kasturba Nagar, Adyar, Chennai, Tamil Nadu-600020
Tel.: 9116687307, 9116687308

SHOLINGANALLUR CAMPUS
S.M. J Tower, Door No.16 IT Expressway, Beside, Novotel Old Mahabalipuram Road Sholinganallur, Chennai-600119, TamilNadu.
Tel : 6366366901 / 6366366902

ANNA NAGAR-WEST CAMPUS
Newry Square, No : 99,13th Main Road, 6th Avenue 1st Block, Anna Nagar West Chennai-600040, Tamil Nadu
Tel : 6366366903 / 6366366904

ALLEN PAAVAI CENTER (TALLENTPRO)
NH-44, Pudhuchethrum, Namakkal, Tamilnadu-607018
Tel.: 9611994455, 9566404272

ALLEN PUDUCHERRY CAMPUSES (TALLENTPRO)

VEDAM CAMPUS (H.O.)
No. 372 & 374, Bharathi Street, Puducherry–605001

GURUGRAM CAMPUS
VIP Nagar, Villianur Main Road, Arumparthapuram, Puducherry–605110

Tel. : 9741303080, 9655995899

ALLEN KOCHI CAMPUSES

PALARIVOTTOM CAMPUS (H.O.)
RK Commercial, PJ Antony cross road, Behind Palarivottam Metro Station, Palarivottom, Kochi, Ernakulum, Kerala.
Tel.: 9116687309, 9116687310

M.G ROAD CAMPUS
Primero Plaza, A K Sheshadiri Road, Near Maharajas College Ground, Kochi, Ernakulum, Kerala.
Tel.: 9116687311, 9116687312

ALLEN TIRUPATI CENTER (TALLENTPRO)
170, First Floor, Above MGB Bajaj,New Balaji Colony, AIR Bypass Road, Tirupati-517501
Tel.: 9900070050

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

BENGALURU (R.O.): "SAMBUDH" 36, 15th Cross, 3rd Block Jayanagar Near Southend Circle, Bengaluru-560011 **Tel : 080-46704000**
Email : bengaluru@allen.ac.in | **Web :** www.allen.ac.in/bengaluru

Corporate Office: "SANKALP", CP-6, Indra Vihar Kota (Raj.), India, 324005 **Tel.:** 0744-2757575
Email: info@allen.ac.in **Web:** www.allen.ac.in

SOME USEFUL CONSTANTS

Atomic No. H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17,Br = 35, Xe = 54, Ce = 58,

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23}\ JK^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8\ ms^{-1}$
- **Stefan–Boltzmann constant** $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3}\ m\text{–}K$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34}\ J\text{–}s$

ENTHUSIAST SRG TEST SERIES

TEST - 11 (PAPER 1)	DATE : 12 - 01 - 2022

SYLLABUS

PHYSICS	:	FULL SYLLABUS
CHEMISTRY	:	FULL SYLLABUS
MATHS	:	FULL SYLLABUS

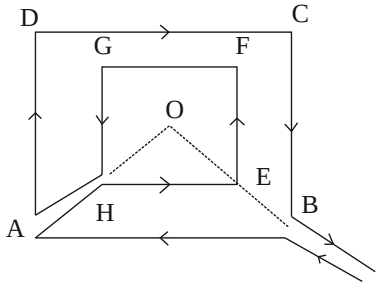
PART-1 : PHYSICS

SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

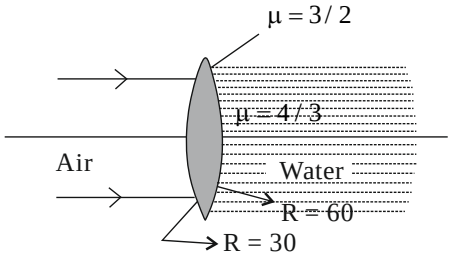
1. ABCD is a square of side 4R and EFGH is a square loop of side 2R. The loop carries a current I. The magnetic field at the centre O of the loop



(A) $\frac{\mu_0 I}{2\pi R}\odot$ (B) $\frac{\mu_0 I}{2\pi R}\otimes$

(C) $\frac{\mu_0 I}{\sqrt{2}\pi R}\otimes$ (D) $\frac{\mu_0 I}{\sqrt{2}\pi R}\odot$

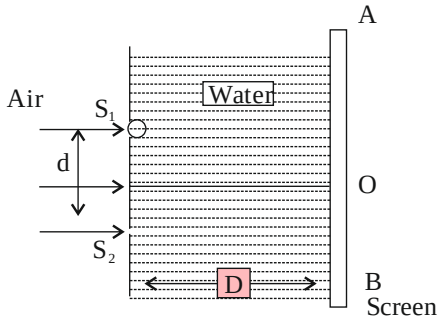
2. A thin lens with radius of curvature 30 cm and 60 cm is placed with water on right side and air on left side, then the focal length of the lens for the parallel incident rays shown in the figure, is



(A) 72 cm (B) 144 cm

(C) 36 cm (D) None of these

3. Young's double slit experiment is submerged in water as shown in the figure. If an air bubble is formed on slit S_1 , then the fringe pattern



(A) Will shift towards A (B) Will shift towards B

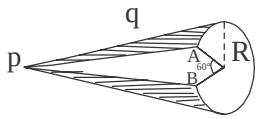
(C) Will not shift (D) Can't say

4. Magnetic field exist in the space and given as $\vec{B} = -\frac{B_0}{l^2}x^2\hat{k}$ where B_0 *and* l are positive constants. A particle having positive charge 'q' and mass 'm' is projected with speed 'v_0' along positive x-axis from the origin. What is the maximum distance of the charged particle from the y-axis before it turns back due to the magnetic field. (Ignore any interaction other than magnetic field)

(A) $\left(\frac{ml^2v_0}{3qB_0}\right)^{1/3}$ (B) $\left(\frac{3ml^2v_0}{2qB_0}\right)^{1/3}$

(C) $\left(\frac{3ml^2v_0}{qB_0}\right)^{1/3}$ (D) $\left(\frac{ml^2v_0}{qB_0}\right)^{1/3}$

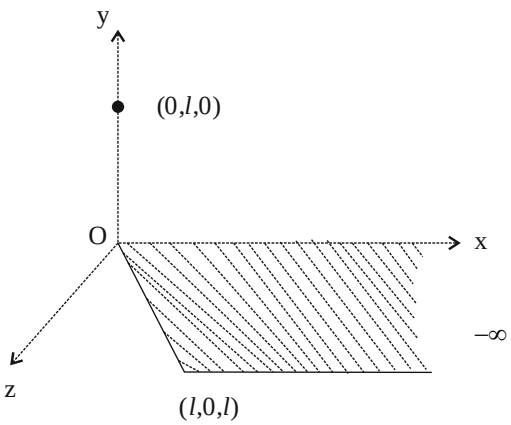
5. A non-conducting hollow cone has charge density σ. A part ABP is cut and removed from the cone. The potential due to the remaining portion of the cone at point 'P' is



(A) $\frac{5}{6}\frac{\sigma R}{\varepsilon_0}$ (B) $\frac{5}{24}\frac{\sigma R}{\varepsilon_0}$

(C) $\frac{5}{3}\frac{\sigma R}{\varepsilon_0}$ (D) $\frac{5}{12}\frac{\sigma R}{\varepsilon_0}$

6. A point charge 'q' is placed at a distance '*l*' (on the y-axis) vertically above the surface which lies in the xz plane as shown in the figure. The flux of electric field passing through the plate is



(A) $\frac{q}{6\varepsilon_0}$ (B) $\frac{q}{18\varepsilon_0}$

(C) $\frac{q}{24\varepsilon_0}$ (D) $\frac{q}{48\varepsilon_0}$

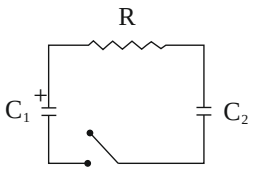
SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

7. In the circuit shown in the figure $C_1 = C_2 = C$ and capacitor C_1 is having initial charge Q. The switch is closed at t = 0. Which of the following options are correct



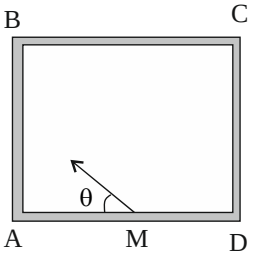
(A) The current at time t through the circuit will be $\left(\frac{Q}{RC}\right)e^{-2t/RC}$

(B) The charge on capacitor C_1 at time t will be $\frac{Q}{2}\left(1+e^{-2t/RC}\right)$

(C) The charge on the capacitor C_2 at time t will be $\frac{Q}{2}\left(1-e^{-2t/RC}\right)$

(D) Total heat loss in the process of charge transfer will be $\frac{Q^2}{8C}$

8. Four identical mirrors are made to stand vertically to form a square arrangement as shown in the top view. A ray starts from the midpoint M of mirror AD and after two reflections reached corner D. Then, angle θ must be



(A) $\tan^{-1}(0.75)$ (B) $\cot^{-1}(0.75)$

(C) $\sin^{-1}(0.75)$ (D) $\cos^{-1}(0.75)$

9. In YDSE, intensity of light wave passing through slits S_1 and S_2 are I_1 and I_2 respectively as shown in figure (1) . The pattern of intensity distribution on the screen is given in figure (2) where x is distance measured from the point O. In this case which of the following statement(s) is / are true
(Given d <<D)

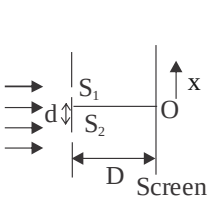


Figure 1

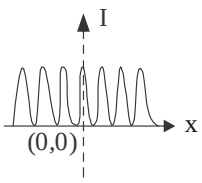
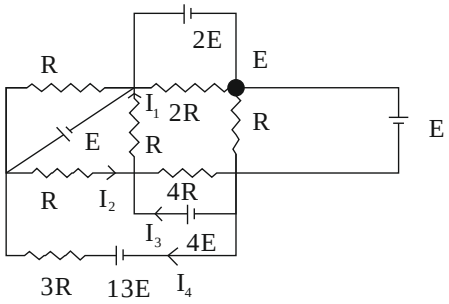


Figure 2

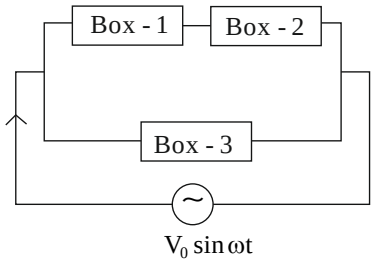
(A) The intensity of light passing through S_1 and S_2 is the same
(B) The phase difference between the light waves passing through S_1 and S_2 to a fixed point on the screen is independent of time
(C) The phase difference between the light waves passing through S_1 and S_2 to a fixed point on the screen varies with time
(D) Light waves passing through S_1 and S_2 have the same wave length

10. In the circuit shown in the figure I_1, I_2, I_3, I_4 denote current in the branches. Choose the **INCORRECT** option



(A) $I_1 > I_2 > I_3 > I_4$
(B) $I_1 = I_2 > I_3 > I_4$
(C) $I_1 < I_3 < I_2 < I_4$
(D) $I_2 < I_1 < I_3 < I_4$

11. In the figure Box-1 contains one capacitor and one resistor in series

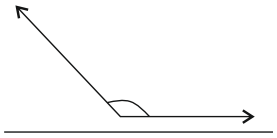


Box - 2 contains one inductor and one resistor in series
R_1 = resistance in Box 1
R_2 = resistance in Box 2

Power factor of Box 1 $= \frac{1}{\sqrt{5}}$

Power factor of Box 2 $= \frac{1}{\sqrt{5}}$

Power factor of (Box 1 + Box 2) $= \frac{1}{\sqrt{2}}$



Power factor of Box 3 $= \frac{1}{\sqrt{2}}$

Resistance of Box 3 = $R_1 + R_2$

Current $I = I_0 \sin(\omega t + \phi)$

On the basis of above mentioned information pick the correct option(s) from below :

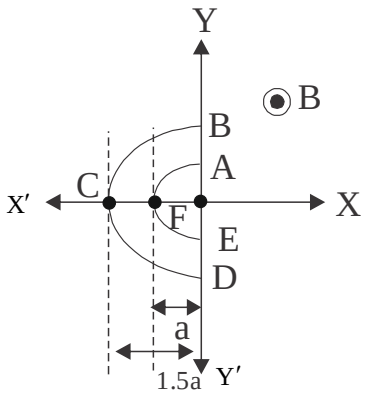
(A) $\frac{R_1}{R_2} = 3$

(B) $Z_3 = 4\sqrt{2}R_2$ (Impedance of Box 3)

(C) $I = \frac{V_0}{4R_2}\sin(\omega t)$

(D) $\phi = (2n+1)\pi$, where n may be some integer

12. In the figure ABCDEFA is a closed conducting wire loop. The part AFE is a portion of an ellipse $\frac{X^2}{4a^2} + \frac{Y^2}{a^2} = 1$ and BCD is a portion of the ellipse $\frac{X^2}{9a^2} + \frac{Y^2}{4a^2} = 1$. A uniform magnetic field $\vec{B} = B_0\hat{k}$ exists in this region. If the frame is rotated about the y-axis with a constant angular velocity 'ω'. The maximum emf induced in the loop is ,



(A) $\frac{\pi\omega B_0 a^2}{2}$

(B) $\pi\omega B_0 a^2$

(C) $2\pi\omega B_0 a^2$

(D) $1.5\pi\omega B_0 a^2$

SECTION–II: (Maximum Marks : 18)

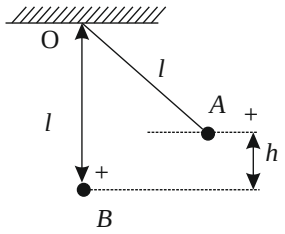
- This section contains **SIX** questions.
- The answer to each question is a **SINGLE / THREE DIGIT INTEGER** ranging from 0 to 999, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

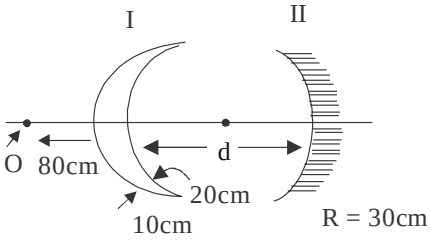
---

13. Wavelength of the k_α characteristics X-ray of iron and potassium are 1.931×10^{-8}cm and 3.737×10^{-8}cm respectively. The atomic number (approximately) of the element for which characteristic k_α wavelength is 2.289×10^{-8}cm is k, the value of 10k is
(atomic number of iron is 26 and atomic number is potassium is 19)

14. A charged particle A hangs with an inextensible string of length *l*. Another charged particle B is fixed at a distance *l* = 10 m below the point of suspension O as shown in the figure. In the equilibrium the particle A is at a height h = 3m above the level of particle B. If the mass of the particle A is m = 2kg and acceleration due to gravity is $g=10m/s^2$, then the electrical potential energy of interaction of the particle in the equilibrium will be



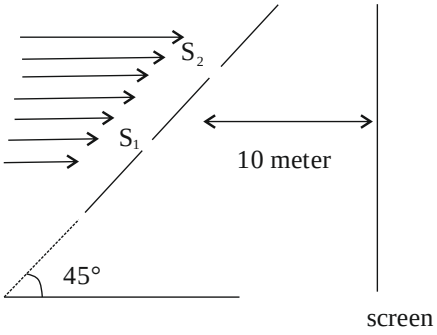
15. If final image formed after two refractions through the lens and one reflection from the mirror forming at same point o then find maximum value of 'd'. (if μ is refractive index of material of lens is 1.5)



16. A ray of light falling on a glass sphere of refractive index $\mu = \sqrt{3}$ such that the directions of the incident ray and emergent ray when produced meet the surface at the same point. The value of angle of incidence is θ. Calculate the value of 2θ in degrees.

17. A charged particle having charge q and mass m is projected into a region of uniform electric field of strength $\vec{E}_0$, with velocity $\vec{V}_0$ perpendicular to $\vec{E}_0$. Throughout the motion apart from electric force particle also experiences a dissipative force of constant magnitude qE_0 and directed opposite to its velocity. If $\left|\vec{V}_0\right| = 6\text{m/s}$, then its speed when it has turned through an angle of 90°, is equal to

18. Consider a plane inclined at an angle 45° with the horizontal has two slits (S_1 and S_2) separated by a distance $d = \sqrt{2}\text{mm}$. The screen is placed at a distance of D = 10 m. A parallel monochromatic light beam of wavelength 5000 Å is incident on the slits as shown. If the fringe width of interference pattern on the screen is $k \times 10^{-3}$ meter, then find the value of k.



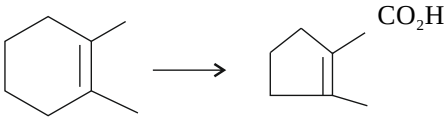
PART-2 : CHEMISTRY

SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. 

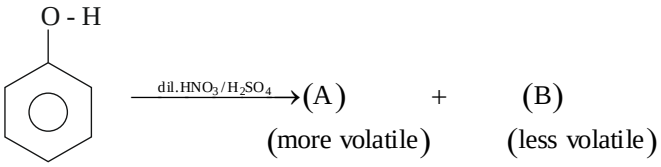
To carry out above conversion, arrange the following reagents in correct order

$O_3 / Zn, \Delta$ $\quad$ $dil.OH^{\ominus} / \Delta$ $\quad$ $NaOCl$ $\quad$ H^+

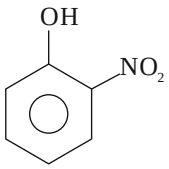
(1) $\quad$ (2) $\quad$ (3) $\quad$ (4)

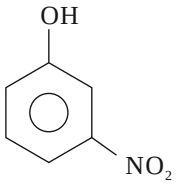
(A) $1 \rightarrow 3 \rightarrow 2 \rightarrow 4$ $\quad$ (B) $1 \rightarrow 2 \rightarrow 4 \rightarrow 3$

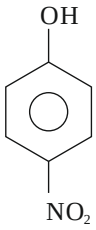
(C) $1 \rightarrow 3 \rightarrow 4 \rightarrow 2$ $\quad$ (D) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

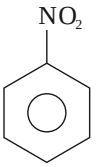
2. 

Product (A) of the above reaction is :

(A) 

(B) 

(C) 

(D) 

3. In which of the following complexes bond order of $C-O$ bond is minimum ?

(A) $\left[Mn(CO)_6\right]^+$ $\quad$ (B) $\left[Cr(CO)_6\right]$

(C) $\left[Ti(CO)_6\right]^{2-}$ $\quad$ (D) $\left[V(CO)_6\right]^-$

4. Identify M and N in the following reaction.

Copper glance $\xrightarrow[\text{in presence of air}]{\text{controlled heating}} M + SO_2$

$M \xrightarrow{N} Cu + SO_2 \uparrow$

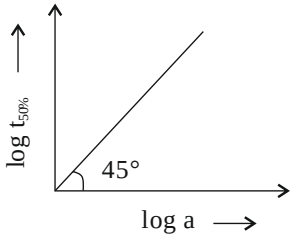
(A) $M = Cu_2O$; $N =$ Self reduction

(B) $M = Cu_2O + Cu_2S$; $N =$ Only heating

(C) $M = Cu_2O$; $N =$ Carbon reduction

(D) $M = Cu_2O$; $N =$ Electrolytic reduction

5. What will be the order of reaction and rate constant for a chemical change having $\log[t_{50\%}]$ vs log [concentration (a)] curves as ?



(A) 0, 1

(B) 1, 1

(C) 2, 2

(D) 3, 1

6. The cell reaction for the given cell is spontaneous if :

$Pt\ ;\ Cl_2\ (P_1)\ \|Cl^-\ (1M)\ \|\ Cl^-\ (1M)\ |\ Cl_2\ (P_2)\ ,\ Pt$

(A) $P_1 > P_2$

(B) $P_1 < P_2$

(C) $P_1 = P_2$

(D) $P_2 = 1$ atm

SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

7. Which of the following is / are correct for lyophilic aquasols ?

(A) Its surface tension is lower than that of H_2O (B) Its viscosity is higher than that of water

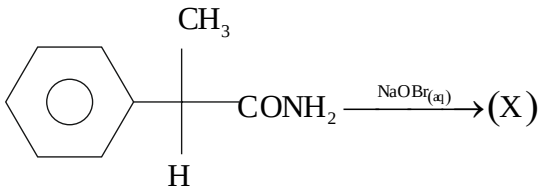
(C) Colloidal silver (D) Colloidal antimony

8. Nitrogen dioxide can be obtained by heating

(A) KNO_3 (B) $Pb(NO_3)_2$

(C) $LiNO_3$ (D) $AgNO_3$

9. Which of the following statements are true about the major product (X) in the reaction ?



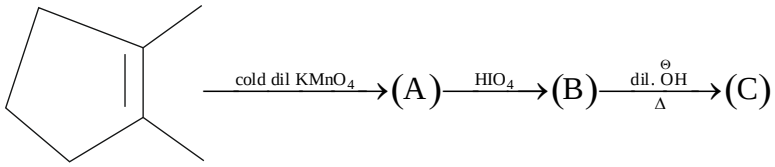
(A) (X) is a cyclic amide

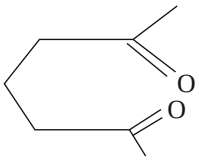
(B) (X) has an asymmetric centre

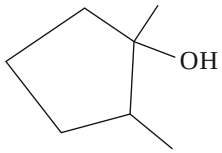
(C) (X) is a primary amine

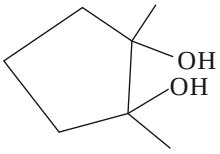
(D) (X) can also be obtained by treatment of NH_3 with $C_6H_5CH(CH_3)Br$

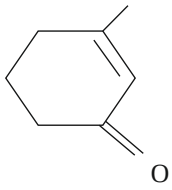
10. In a reaction sequence



(A) A is 

(B) B is 

(C) A is 

(D) C is 

11. Which of the following is / are correct ?

(A) On strong heating $FeCl_3$ gives $FeCl_2$ and Cl_2

(B) Dilute $CuCl_2$ is blue due to formation of $[CuCl_4]^{2-}$

(C) PH_3 can be obtained by heating red P with NaOH

(D) PH_3, H_3PO_3 and H_3PO_2, all are reducing agents

12. $2X(g)+Y(g)+3Z(g) \rightarrow Products$

The rate equation of the above reaction is given by :

$Rate = K[X][Y]^{0}[Z]^{2}$

Choose correct statements :

(A) The rate of reaction becomes doubled by doubling the conc.of (X)

(B) Rate of reaction decreases by reducing the conc of (Y) to half of the original value

(C) The half life of (Z) increases by increasing its conc

(D) If 75% of (X) undergoes reaction in 20 sec, 50% of (X) will react in 10 sec

SECTION–II: (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE / THREE DIGIT INTEGER** ranging from 0 to 999, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 In all other cases.

---

13. Freezing point of 0.2 M KCN (100% ionisation) solution is $-0.7°C$. On adding 0.1 mole of $Hg(CN)_2$ to one litre of the 0.2 M KCN solution, the freezing point of the solution becomes - 0.53°C due to the reaction

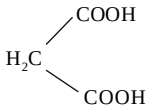
$Hg(CN)_2 + mCN^- \rightarrow Hg(CN)_{m+2}^{m-}$. What is the value of m, assuming molality = molarity ?

14. If $E^0_{Fe^{2+}/Fe} = 5V$ and $E^0_{Fe^{3+}/Fe^{2+}} = 2V$ then $E^0_{Fe^{3+}/Fe}$ will be

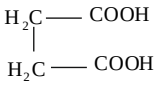
15. A cell made up of two hydrogen electrodes. The positive electrode is in contact with a solution of $10^{-4}M$ hydrogen ions. The emf of the cell is 0.118 V at 25°C. The concentration of hydrogen ions at negative electrode is $10^{-x}M$. Then x = ?

16. When Mg_2C_3 is reacted with water, a compound X is formed which on heating with Na followed by treatment with C_2H_5Br gives another compound Y. Calculate the number of sigma bond in Y.

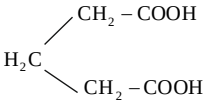
17. How many of the following acids easily undergo decarboxylation on heating ?



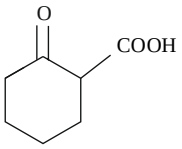
1



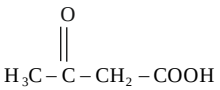
2



3



4



5

18. No.of geometrical isomers for $[M(gly)Cl_2Br_2]^+$ is :

PART-3 : MATHEMATICS

SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. If $f:[0,1]\to[0,\infty)$ is a differentiable function with decreasing first derivative, such that $f(0)=0$ & $f'(x)>0$ then

(A) $\int_0^1 \frac{dx}{(f(x))^2+1} > \frac{f(1)}{f'(1)}$ (B) $\int_0^1 \frac{dx}{(f(x))^2+1} < \frac{f(1)}{f'(1)}$

(C) $\int_0^1 \frac{dx}{(f(x))^2+1} > \frac{\tan^{-1}(f(1))}{f'(1)}$ (D) $\int_0^1 \frac{dx}{(f(x))^2+1} = \frac{f(1)}{f'(1)}$

2. Let $a\in[0,4]$. The maximum area bounded by the curves $y=1-|x-1|$ & $y=|2x-a|$ is

(A) 1/3 (B) 1/4
(C) 1/2 (D) None of these

3. If $f:R\to R$ is a continuous function satisfying $f(0)=1$ and $f(2x)-f(x)=x\ \forall x\in R$ and $\lim_{x\to\infty}\left(f(x)-f\left(\frac{x}{2^n}\right)\right)=p(x)$ then p(x) is

(A) Constant function (B) A linear polynomial in x
(C) A quadratic polynomial in x (D) A cubic polynomial in x

4. If f(x) is continuous function such that $f(x)>0\ \forall x\ge 0$ and $(f(x))^{2014}=1+\int_0^x f(t)dt$ then the value of $(f(2014))^{2013}$ is equal to

(A) 2013 (B) 2014
(C) 2015 (D) 2016

5. Let $f:[0,\infty)\to R$ be a continuous and strictly increasing function such that $f^4(x)=\int_0^x t^2 f^3(t)dt\ \forall x>0$. The area enclosed by y = f(x), the x-axis and the ordinates at x = - 1 and x = 4 is

(A) $\frac{17}{48}$ (B) $\frac{65}{48}$

(C) $\frac{257}{48}$ (D) $\frac{235}{48}$

6. For all $x \in \left[-\frac{7}{2}, \frac{7}{2}\right]$, let $f(x) = \begin{cases} |[x]|, & 0 \le \{x\} < \frac{1}{2} \\ |x|, & \frac{1}{2} \le \{x\} < 1 \end{cases}$

Note : [x] denotes greatest integer less than or equal to x and $\{x\}$ denotes fractional part of x

If L is the number of points of discontinuity and M is the number of points of non-differentiability of the function f(x), then (L + M) is equal to

(A) 18 (B) 20

(C) 21 (D) 22

SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

7. Let f, g & h are functions differentiable on some open interval around origin (0,0) and satisfy the equations $f' = 2f^2gh + \frac{1}{gh}$, $f(0) = 1, g' = fg^2h + \frac{4}{fh}$, $g(0) = 1, h' = 3fgh^2 + \frac{1}{fg}$, $h(0) = 1$.

The function f is given by

(A) $f(x) = 2^{-\frac{1}{12}}\left(\frac{\sin\left(6x + \frac{\pi}{4}\right)}{\cos^2\left(6x + \frac{\pi}{4}\right)}\right)^{1/6}$

(B) $f(x) = 2^{-\frac{1}{6}}\left(\frac{\sin\left(6x + \frac{\pi}{4}\right)}{\cos^2\left(6x + \frac{\pi}{4}\right)}\right)^{1/12}$

(C) $f(x) = (\sec 12x)^{1/12}(\sec 12x + \tan 12x)^{1/4}$

(D) $f(x) = (\sec 12x)^{1/14}(\sec 12x + \tan 12x)^{1/12}$

8. For any real number $\lambda \geq 1$. Let $f(\lambda)$ denote the real solution of the equation $x(1+\ln x)=\lambda$ then

(A) $f(\lambda)$ is strictly increasing function in its domain

(B) $\lim\limits_{\lambda\to\infty} f(\lambda) \to \infty$

(C) If $h(x)=x(1+\ln x), h:[1,\infty)\to[1,\infty)$ then $f(\lambda)$ is inverse function of h(x)

(D) $\lim\limits_{\lambda\to\infty} \dfrac{f(\lambda)\ln\lambda}{\lambda}=1$

9. A point Q moves on the positive y-axis and a point P follows the point Q such that the motion of P is always towards Q and distance between P and Q is always equal to some constant K > 0. If initially Q is at origin and P is at (K,0) then

(A) The differential equation representing the curve traced by P is $\dfrac{dy}{dx}=\dfrac{\sqrt{K^2-x^2}}{-x}$

(B) y-axis is asymptote to the curve traced by P

(C) The equation of curve traced by P is $y=-K\ln\left(\dfrac{K+\sqrt{K^2-x^2}}{x}\right)-\sqrt{K^2-x^2}$

(D) The curve traced by P will have x-axis as asymptote

10. The function $f(x)=x^{\frac{1}{3}}(x-1)$

(A) Has vertical tangent at x = 0

(B) Area enclosed by curve with x-axis from $x=-1$ to $x=1$ is $\dfrac{3}{2}$

(C) Is concave up in $(0,\infty)$

(D) Is decreasing for $x\in\left(-\infty,\dfrac{1}{4}\right)$

11. Let $\vec{r}$ be a position vector of a variable point in cartesian xy-plane such that $\vec{r}.\left(-8\hat{i}+10\hat{j}-\vec{r}\right)=40$ and $M=\max\left|\vec{r}+2\hat{i}-3\hat{j}\right|^2, m=\min\left|\vec{r}+2\hat{i}-3\hat{j}\right|^2$, then (M + m) is less than or equal to

(A) 9 (B) 18

(C) 27 (D) 28

12. A computer producing factory has only two plants T_1 and T_2. Plant T_1 produces 20% and plant T_2 produces 80% of the total computers produced. 7% of computers produced in the factory turn out of be defective. It is known that

P(computer turns out to be defective given that it is produced in plant T_1)

= 10 P (computer turns out to be defective given that it is produced in plant T_2),

where P(E) denotes the probability of an event E. A computer produced in the factory is randomly selected and it does not turn out to be defective. Then the probability that it is produced in plant T_2 is

(A) $\dfrac{36}{73}$ (B) $\dfrac{47}{79}$

(C) $\dfrac{78}{93}$ (D) $\dfrac{75}{83}$

SECTION–II: (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE / THREE DIGIT INTEGER** ranging from 0 to 999, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 In all other cases.

---

13. If $\lim\limits_{r\to\infty}\dfrac{r^C\int\limits_0^{\pi/2}x^r\sin x\,dx}{\int\limits_0^{\pi/2}x^r\cos x\,dx}=L$, then value of $\pi L-C$ is equal to ..... $(C\in R, L>0)$

14. n biased coins , with m^{th} coin having probability of throwing head equal to $\dfrac{1}{(2m+1)}$ (m = 1, 2, ...,n), are tossed once. The probability of getting an odd numbers of heads, if results for each coin are independent, is $\dfrac{\alpha n}{\beta n+\gamma}$ then $\alpha+\beta+\gamma$ is equal to (given α,β & γ are co-prime natural numbers)

15. Let $\vec{A}=\hat{i}-2\hat{j}+3\hat{k},\vec{B}=2\hat{i}+\hat{j}-\hat{k},\vec{C}=\hat{j}+\hat{k}$. If the vector $\vec{B}\times\vec{C}$ can be expressed as a linear combination $\vec{B}\times\vec{C}=x\vec{A}+y\vec{B}+z\vec{C}$ where x, y, z are scalar, then find the value of x + y + z .....

16. Find the sum of an infinite geometric series whose first term is $\lim\limits_{x\to0}\sum\limits_{k=1}^{2013}\dfrac{\left\{\dfrac{x}{\tan x}+2k\right\}}{2013}$

and whose common ratio is the value of $\lim\limits_{x\to0}\dfrac{e^{\tan^3 x}-e^{x^3}}{2\ln\left(1+x^3\sin^2 x\right)}$

[Note : where {k} denotes fractional part of k]

17. If $\overrightarrow{V_1}=\hat{i}-2\hat{j}+3\hat{k}$ and $\overrightarrow{V_2}=a\hat{i}+b\hat{j}+c\hat{k}$ where $a,b,c\in\{-1,0,1,2,3\}$ then find the number of non-zero vectors $\overrightarrow{V_2}$ which are normal to $\overrightarrow{V_1}$

18. Let $A=\begin{pmatrix}2x\\5\\-3x^2\end{pmatrix},B=\begin{pmatrix}2a & -b & 5c\end{pmatrix},C=\begin{pmatrix}5x & -1 & 4x^2\end{pmatrix}$ and $D=\begin{pmatrix}2\\1\\4\end{pmatrix}$ be four given matrices, where a, b, c and $x\in R$.

Given that $t_r(AB)=t_r(DC)\ \forall x\in R$ where $t_r(A)$ denotes trace of matrix A, then if

$\overrightarrow{V_1}=a\hat{i}+b\hat{j}+c\hat{k},\overrightarrow{V_2}=\dfrac{2\hat{i}}{5}+10\hat{j}-30\hat{k}$ and $\overrightarrow{V_3}=\hat{k}$ then find the value of $\left[\left(\overrightarrow{V_1}\times\overrightarrow{V_2}\right).\overrightarrow{V_3}\right]^2$

[Note : [k] denotes greatest integer less than or equal to k]

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 – 2022)

TARGET : JEE (M + A) : 2022

ENTHUSIAST SRG TEST SERIES

Test Type: ONLINE **Test Date : 12 - 01 - 2022** **Pattern:** JEE advance

Time : 3 Hours **TEST NO - 11 (PAPER 2)** **Maximum Marks : 186**

Instructions

READ THE INSTRUCTIONS CAREFULLY

QUESTION PAPER FORMAT AND MARKING SCHEME

1. The question paper has **three parts** : **PHYSICS**, **CHEMISTRY** and **MATHEMATICS**. Each part has **THREE SECTIONS**.
2. Carefully read the instructions given at the beginning of each section.
3. **Section I(i):** This section contains **FOUR** questions. The answer to each question is a **ONLY ONE CORRECT OPTION.**

   Marking scheme : +3 for correct answer, 0 if not attempted and –1 in all other cases.
4. **Section I(ii):** This section contains **EIGHT** questions. The Answer to each question is a **ONE OR MORE THAN ONE CORRECT OPTIONS**.

Marking scheme :	+4	for correct answer, 0 if not attempted and –1 in all other cases.
Partial Marks :	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks :	+2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks :	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.

5. **Section II:** This section contains **SIX** questions. The answer to each question is a **SINGLE OR THREE DIGIT INTEGER**.

   Marking scheme : +3 correct answer, 0 if not attempted and 0 in all other cases.

ALLEN SOUTH INDIA CENTERS

ALLEN BENGALURU CAMPUSES

JAYANAGAR | KORAMANGALA | MARATHAHALLI | BANASWADI | HEBBAL | HSR LAYOUT
BANNERGHATTA| BASAVESHWARA NAGAR | SARJAPURA | JALAHALLI | INDIRANAGAR
WHITEFIELD

Tel : 080-46704000

ALLEN MYSURU CENTER

SARASWATHIPURAM CAMPUS
B M. Arcade, No-2923, 1st Main, 5th Cross, Saraswathipuram, Opp. SVC Bank, Mysuru-570009, Karnataka
Tel. 0821-4526818, 9945588588

ALLEN MANGALURU CENTER : (TALLENTPRO)
Vikaas Pre-University College Vikas Group of institutions, Airport Road, Mary Hill Mangaluru, Karnataka-575008
Tel.: 9900090058

ALLEN CHENNAI CAMPUSES

ASHOK NAGAR CAMPUS (H.O.)
No. 93 & 94, Next to Kasi Theatre, Jafferkhanpettai, Ashok Nagar, Chennai-600083, Tamil Nadu.
Tel.: 044-40261000

VELACHERY CAMPUS
Plot No.13, New survey No - 347/2A, Vijaya Nagar, 1st Cross, Main Road, Velachery, Chennai - 600042, Tamil Nadu
Tel: 9116687303, 9116687304

ANNA NAGAR-EAST CAMPUS
New No : 80, Old no : 419, Kilpauk Garden Road, Anna Nagar East, Chennai–600010, Tamil Nadu
Tel.: 9116687305, 9116687306

ADYAR CAMPUS
Above Nalli Silks,City Towers, 7, 3rd Cross St, Kasturba Nagar, Adyar, Chennai, Tamil Nadu-600020
Tel.: 9116687307, 9116687308

SHOLINGANALLUR CAMPUS
S.M. J Tower, Door No.16 IT Expressway, Beside, Novotel Old Mahabalipuram Road Sholinganallur, Chennai-600119, TamilNadu.
Tel : 6366366901 / 6366366902

ANNA NAGAR-WEST CAMPUS
Newry Square, No : 99,13th Main Road, 6th Avenue 1st Block, Anna Nagar West Chennai-600040, Tamil Nadu
Tel : 6366366903 / 6366366904

ALLEN PAAVAI CENTER (TALLENTPRO)
NH-44, Pudhuchethrum, Namakkal, Tamilnadu-607018
Tel.: 9611994455, 9566404272

ALLEN PUDUCHERRY CAMPUSES (TALLENTPRO)

VEDAM CAMPUS (H.O.)
No. 372 & 374, Bharathi Street, Puducherry–605001

GURUGRAM CAMPUS
VIP Nagar, Villianur Main Road, Arumparthapuram, Puducherry–605110

Tel. : 9741303080, 9655995899

ALLEN KOCHI CAMPUSES

PALARIVOTTOM CAMPUS (H.O.)
RK Commercial, PJ Antony cross road, Behind Palarivottam Metro Station, Palarivottom, Kochi, Ernakulum, Kerala.
Tel.: 9116687309, 9116687310

M.G ROAD CAMPUS
Primero Plaza, A K Sheshadiri Road, Near Maharajas College Ground, Kochi, Ernakulum, Kerala.
Tel.: 9116687311, 9116687312

ALLEN TIRUPATI CENTER (TALLENTPRO)
170, First Floor, Above MGB Bajaj,New Balaji Colony, AIR Bypass Road, Tirupati-517501
Tel.: 9900070050

SOME USEFUL CONSTANTS	
Atomic No.	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17,Br = 35, Xe = 54, Ce = 58,
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23}\ JK^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8\ ms^{-1}$
- **Stefan–Boltzmann constant** $\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3}\ m\text{–}K$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34}\ J\text{–}s$

ENTHUSIAST SRG TEST SERIES

TEST - 11 (PAPER 2)	DATE : 12 - 01 - 2022

SYLLABUS

PHYSICS	:	FULL SYLLABUS
CHEMISTRY	:	FULL SYLLABUS
MATHS	:	FULL SYLLABUS

PART-1 : PHYSICS

SECTION–I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. You are holding a bottle of bubbled water inside a car moving forward. When the driver applies the brakes:

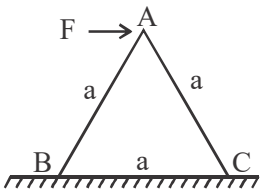
(A) Bubbles in the middle of the liquid will start to move forward with respect to the bottle.

(B) Bubbles will start to move backward with respect to the bottle.

(C) Bubbles will stay at the same horizontal location in the water

(D) Depending on the speed of the car, bubbles might move forward or backward

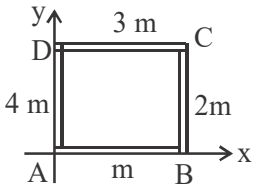
2. A horizontal force F is applied at the top of an equilateral triangular block having mass m. The minimum coefficient of friction required to topple the block before translation will be



(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{3}}$ (D) None of these

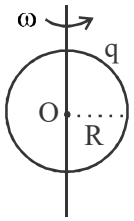
3. Four rods of equal lengths AB, BC, CD and DA have mass m, 2m, 3m and 4m respectively are place in x-y plane (see figure) given AB = BC = CD = DA = 4 meters. The centre of mass of the system will satisfy (x and y are in meters)



(A) $x > 2, y > 2$ (B) $x > 2, y < 2$

(C) $x < 2, y < 2$ (D) $x < 2, y > 2$

4. A hollow non conducting sphere of radius R has a positive charge q uniformly distributed on its surface. The sphere starts rotating with a constant angular velocity ω about an axis passing through center of sphere, as shown in the figure. Then the net magnetic field at center of the sphere is



(A) $\frac{\mu_0 q\omega}{2\pi R}$ (B) $\frac{\mu_0 q\omega}{3\pi R}$

(C) $\frac{\mu_0 q\omega}{6\pi R}$ (D) $\frac{\mu_0 q\omega}{12\pi R}$

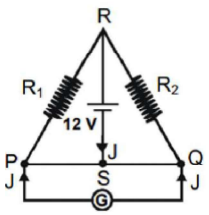
SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

5. In the circuit shown $R_1 = R_2 = 10\,\Omega$ and resistance per unit length of wire PQ = 1 Ω /cm and length PQ = 10 cm. If R_2 is made $20\,\Omega$. Then to get zero deflection in galvanometer. S is midpoint of wire PQ



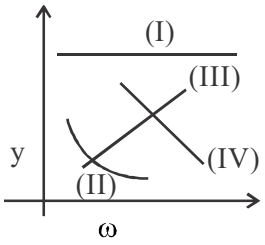
(A) The jockey at P can be moved towards right 2 cm

(B) The jockey at Q can be moved towards left 2cm

(C) The jockey at S can be moved towards left a distance 5/3 m

(D) The jockey at all position fixed and R_1 should be made 20Ω

6. In a series LCR circuit, different physical quantities vary with frequency ω. Which of the following curves represent correct frequency variation of the corresponding quantity?



(A) Curve I for R
(B) Curve II for current I
(C) Curve III for X_L
(D) Curve IV for X_c

7. An observer 'O' at an accelerated trolley observes a projectile Q of mass 5 kg moving under uniform gravitational field of earth acting towards negative z-axis. He observes the momentum $\vec{p}$ of the projectile Q as $\vec{P} = 15t\hat{i} - 10t^2\hat{j} - 50t\hat{k}$ (kg m/s), where t denotes time in second. Find the acceleration of the trolley at t = 2 sec. (Take g = 10 m/s^2)

(A) $6\hat{i} - 8\hat{j} - 20\hat{k}\,m/s^2$
(B) $3\hat{i} - 8\hat{j}\,m/s^2$
(C) $-3\hat{i} - 8\hat{j}\,m/s^2$
(D) $-3\hat{i} + 8\hat{j}\,m/s^2$

8. Pick the correct option
(A) Elastic deformation energy is basically electrostatic potential energy.
(B) In general the angle of contact of a solid liquid pair decreases with rise in temperature.
(C) Siphon action can not be explained by capillary rise phenomenon.
(D) An ideal fluid must be transparent.

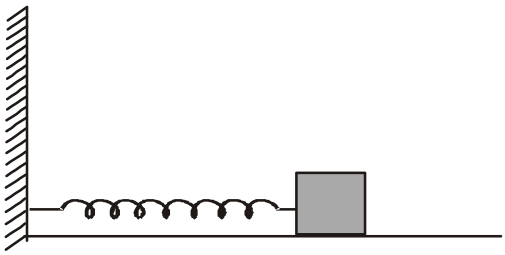
9. X-rays from a tube with a target A of atomic number Z shows strong K line for target A and weak K lines for impurities. The wavelength of K_α lines is λ_z for target A and λ_1 and λ_2 for two impurities.

If $\frac{\lambda_z}{\lambda_1} = 4$ and $\frac{\lambda_z}{\lambda_2} = \frac{1}{4}$ and taking screening constant of K_α lines to be unity, select the correct statement(s)

(A) The atomic number of first impurity is 2Z – 1
(B) The atomic number of first impurity is 2Z + 1
(C) The atomic number of second impurity is $\frac{(Z+1)}{2}$
(D) The atomic number of second impurity is $\frac{Z}{2} + 1$

10. An equi-convex lens (μ = 1.5) is combined with an equi concave lens (μ = 1.3). The radii of curvature of all surfaces is the same. (Assume thin lenses)
(A) The combination behaves like a converging lens in medium of refractive index 1.4 and diverging lens in a medium of refractive index 1.6.
(B) The combination behaves like a converging lens in a medium of refractive index 1.2 as well as 1.4.
(C) The combination behaves like a diverging lens in medium of refractive index 1.2 as well as 1.6.
(D) The combination behaves like a diverging lens in medium of refractive index 1.4 and converging lens in a medium of refractive index 1.6.

11. A block of mass 1 gm carrying a charge 1 μC is connected to a spring of stiffness $\left(\pi^2 \times 10^{-3}\right)$ N/m whose other end is fixed to a wall. The block rests on a smooth ground as shown. By making a suitable arrangement, an electric field lying in the plane of the figure, directed towards right with strength $\left(\frac{\pi}{10}\right)$ MV / m is switched on exactly for an interval of 1 ms after every half second. Take the moment when the field is switched on for first time as t = 0. Now, select the correct options.



(A) The maximum extension in the spring is 0.1414 m (nearly)

(B) The spring never gets compressed.

(C) The time taken by the block to start from mean position and reach the position of maximum elongation of spring is 0.75s.

(D) The time taken by the block to start from mean position to the position of maximum elongation of spring is 0.5 s.

12. In a "Positronium" atom a pair of electron and positron orbit each other. If Bohr's atomic model is valid, then in comparison with Bohr's atomic hydrogen,
(assume that mass of proton >>> mass of electron)
(A) The Rydberg's constant for positronium is half
(B) The ionisation energy for positronium is double
(C) The radius of positronium is double
(D) The kinetic energy of electron in the center of mass frame of positronium is one fourth that of the electron wrt proton in the H-atom

SECTION–II: (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE / THREE DIGIT INTEGER** ranging from 0 to 999, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

---

13. A circular disc of radius R carries surface charge density $\sigma(r)=\sigma_0\left(1-\frac{r}{R}\right)$ where σ_0 is a constant and r is the distance from the center of the disc. Electric flux through a large spherical surface that encloses the charged disc completely is ϕ_0. Electric flux through another spherical surface of radius $\frac{R}{4}$ and concentric with the disc is ϕ. Then the ratio $\frac{\phi_0}{\phi}$ is_____

14. A liquid at 30°C is poured very slowly into a Calorimeter that is at temperature of 110°C. The boiling temperature of the liquid is 80°C. It is found that the first 5 gm of the liquid completely evaporates. After pouring another 80 gm of the liquid the equilibrium temperature is found to be 50°C. The ratio of the Latent heat of the liquid to its specific heat will be _____°C.
[Neglect the heat exchange with surrounding]

15. The pitch of a screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line and zero of main scale is not visible. When a plate is placed between the studs, the main scale reads 5 divisions and the circular scale reads 25 divisions. The true thickness of plate is $a\times10^{-5}$ m, find the value of a.

16. A source of sound and an observer are moving with constant velocities of 10 m/s and 60 m/s respectively. The source emits sound of frequency 657 Hz. At a certain moment, the distance between the source and observer is 415 m and the observer is moving in a direction making angle 37° with direction of position vector of source with respect to observer while the source is headed directly towards the observer. If the frequency of sound (in Hz) perceived by the observer at an instant exactly 5 seconds after this moment is x, then find the value of $\frac{x}{225}$. The speed of sound is 225 m/s. cos 37° = 4/5.

17. Given that mass of Earth is M and that of moon is m and distance between them is R. Neglect effect of any other planet or sun. A small mass is projected from midway from Earth and moon its escape velocity is $P\sqrt{\frac{G}{R}(m+M)}$ find P.

18. A plane thick wall having uniform surface temperature along planes are 0°K and T_0°K ($T_0 = 300$K) at x = 0 and x = x_0 respectively. Thermal conductivity varies linearly with temperature $K = K_0(1 + T)$. The temperature of wall at the plane $x = 2x_0$ is T' kelvin then find the value of $\frac{T'^2+2T'}{200}$

PART-2 : CHEMISTRY

SECTION–I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. In a CCP crystal of edge length 'a' formed by an anion having radius r_-, assuming no distortion in the unit cell, the ratio of radius of tetrahedral void to octahedral void $\left(\frac{r_t}{r_o}\right)$ is

(A) $\frac{\sqrt{6}-2}{\sqrt{8}-2}$ (B) $\frac{\sqrt{3}-2}{\sqrt{8}-2}$

(C) $\frac{\sqrt{6}-2}{\sqrt{2}-2}$ (D) $\frac{\sqrt{3}-2}{\sqrt{2}-2}$

2. The temperature coefficient of standard cadmium cell is $5.0\times10^{-5}\,VK^{-1}$ whose emf at 27°C is 1.015 V. During the cell operation, the temperature will ___ (The number of electrons in overall process involved are two).

(A) Increase (B) Decrease

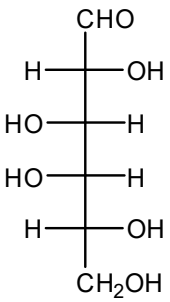
(C) Either (D) Remains constant

3. In the extraction of copper, the metal formed in the Bessemer's converter is due to the reaction :

(A) $Cu_2S \longrightarrow 2Cu + S$ (B) $2Cu_2O \longrightarrow 4Cu + O_2$

(C) $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$ (D) $2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$

4. For the carbohydrate given below, which is not a true statement :



(A) On reaction with Conc.HNO_3, it gives an optically inactive compound

(B) On reduction with $LiAlH_4$, it gives a meso compound

(C) Both L-Glucose and the above molecule will form same osazone

(D) It is a C_4 epimer of D (+) - Glucose

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

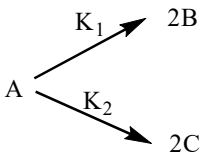
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

5. 'Spin only' magnetic moment of Ni in $[Ni(dmg)_2]$ is same as that found in

(A) Ni in $[NiCl_2(PPh_3)_2]$ (B) Mn in $[MnO_4]^-$

(C) Co in $[CoBr_4]^{2-}$ (D) Pt in $[Pt(H_2O)_2Br_2]$

6. For the reaction



Where $K_1 = 6.93 \times 10^{-3}$ min^{-1}

$K_2 = 1.386 \times 10^{-2}$ min^{-1}

Initially reaction is started with 2 M of A. If both A, B, and C are optically active and have an optical rotation per unit concentration of +40°, –50° and +24° respectively, then which of the following statement(s) is/are true

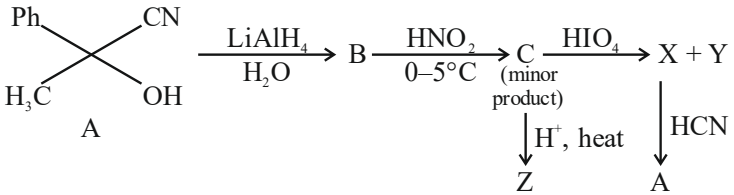
(A) The reaction has a half life of 45 min

(B) When 50% of A is consumed, the reaction mixture has an optical rotation of +38.67°

(C) When 75% of the A is consumed, the reaction mixture has 0° optical rotation

(D) At infinite time, the reaction mixture is levo rotatory

7.



For the above sequence of reactions, which of the following statement(s) is/are true?

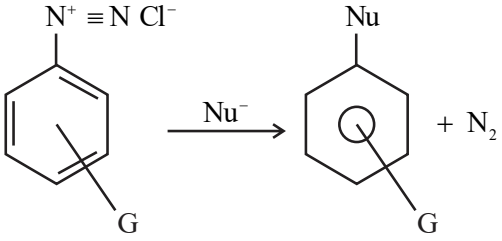
(A) X gives yellow precipitate with I_2 in presence of NaOH

(B) Y is reduced by Fehling's solution

(C) Z gives silver mirror with $[Ag(NH_3)_2]^+$

(D) X reduces Tollen's reagent.

8.



Which of the following is/are not correct regarding above reaction ?

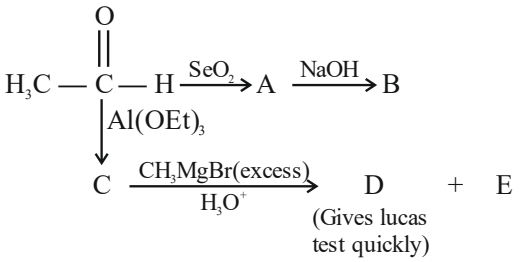
(A) If 'G' is a +R group and present at para position, it will increase the rate of reaction

(B) If 'G' is a –R group and present at para position, it will increase the rate of reaction

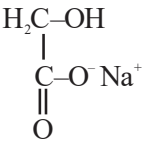
(C) If 'G' is +I group and present at meta position, it will increase the rate of reaction

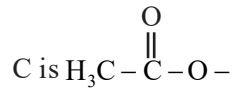
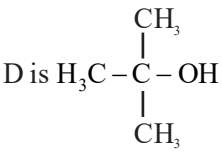
(D) If 'G' is –I group and present at meta position, it will increase the rate of reaction

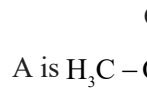
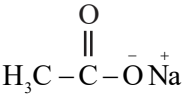
9. In the given reaction sequence

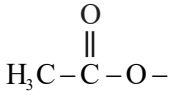
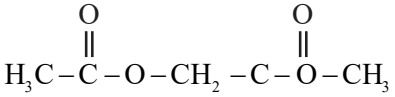


Which of the following is correct?

(A) A is  B is 

(B) C is  ; D is 

(C) A is  ; B is 

(D) A is  ; C is 

10. Which of the following is/are incorrect statement for $KMnO_4$

(A) It can be prepared by the oxidation of K_2MnO_4 by $H_2S_2O_8$

(B) On heating in acidic medium it disproportionates to give MnO_2 and MnO_4^{-2}

(C) As an oxidising agent in neutral medium, $KMnO_4$ equivalent mass is $\frac{M}{3}$

(D) It can be used as primary standard solution in standardisation of oxalic acid sample in HCl medium.

11. For the reaction $P_4 + SOCl_2 \rightarrow A + B + SO_2$, which of the following statements is/are false

(A) If A is phosphorous compound, then on hydrolysis it gives H_3PO_4

(B) If B is a compound of S, then is has similar structure that of H_2O_2

(C) In the above reaction, S undergoes disproportionation

(D) A in solid state exists as $[PCl_4]^+[PCl_6]^-$

12. The ionic conductance at infinite dilution of Al^{+3} and SO_4^{-2} ions are 60 and 80 $\Omega^{-1}cm^2eq^{-1}$, respectively. The correct details regarding $Al_2(SO_4)_3$ will be

(A) The molar conductance is 140 $\Omega^{-1}cm^2mol^{-1}$

(B) The equivalent conductance is 140 $\Omega^{-1}cm^2mol^{-1}$

(C) The molar conductance is 840 $\Omega^{-1}cm^2mol^{-1}$

(D) The molar conductance is 23.33 $\Omega^{-1}cm^2mol^{-1}$

SECTION–II: (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE / THREE DIGIT INTEGER** ranging from 0 to 999, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

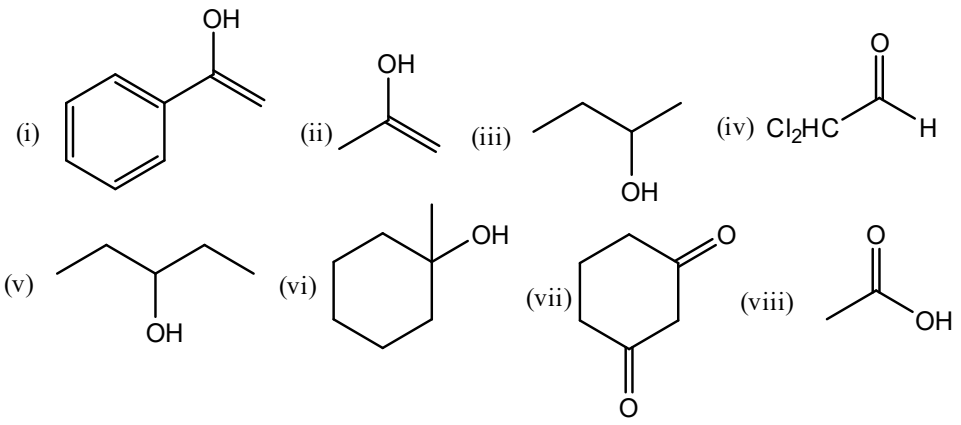
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 In all other cases.

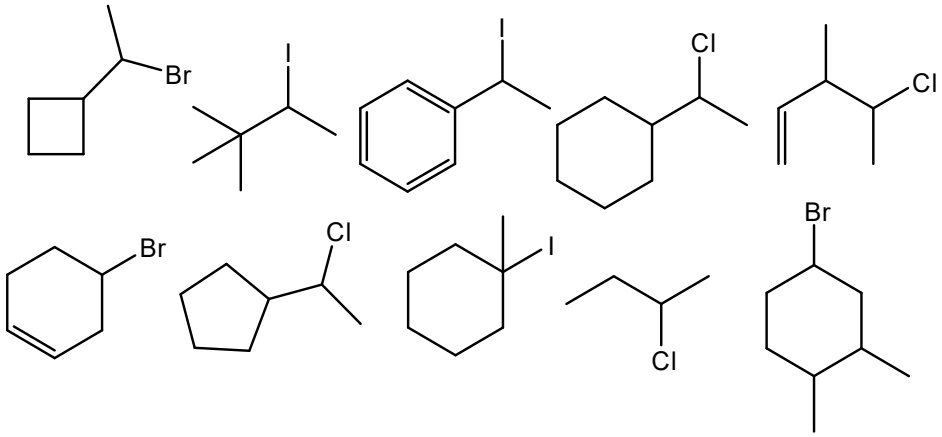
13. At 10°C, the osmotic pressure of urea solution is 500 mm. The solution is diluted and the temperature is raised to 25°C, when the osmotic pressure is found to be 105.3 mm. Then the extent of dilution is

14. A metal cube of edge length 'a' cm is cut n number of times. It is cut such that, per each cut, each cube is divided into 8 smaller cubes of same edge length. Then what should be the value of n, if the final surface area becomes 1048576 times of initial surface area

15. How many of the following molecules does not give positive haloform test.



16. How many of the following substrates will show rearrangement during S_N1 reaction?



17. For the coagulation on 200 mL of As_2S_3 solution, 10 mL of 1 M NaCl is required. What is the coagulating value of NaCl.

18. Number of possible isomers that can be formed (including stereo) for an octahedral complex containing Co^{+3} metal cation, two 'en' ligands, two NO_2^{-1} and one NO_3^{-1} ions is X. Then $\frac{X}{3}$ will be

PART-3 : MATHEMATICS

SECTION–I(i) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{7n}\left(\dfrac{r^6+6r^5+5r^4+4r^3+3r^2+2r+1}{7n^7+6n^6+5n^5+4n^4+3n^3+2n^2+7n+1}\right)$, $k \in R$, $n \in N$ is

(A) 6^7 (B) 6^5 (C) 5^7 (D) 7^5

2. If two points P & Q on the hyperbola $9x^2-5y^2=1$ whose centre is C be such that CP is perpendicular to CQ, then $\dfrac{1}{CP^2}+\dfrac{1}{CQ^2}$ is

(A) Equal to the number of integers for which n^4-20n^2+4 is a prime number
(B) Equal to the twice the number of integers for which n^4-20n^2+4 is a prime number
(C) Equal to 4
(D) Is equal to 2

3. The value of $S=\sec11°\sec19°-2\cot71°$ equals

(A) $2\cot11°$ (B) $\tan19°$
(C) $2\tan11°$ (D) $\dfrac{1}{2}\tan19°$

4. Let r_1, r_2, r_3, r_4 be the four roots of the polynomial $x^4-4x^3+8x^2-7x+3$. Find the absolute value of $\dfrac{r_1^2}{r_2^2+r_3^2+r_4^2}+\dfrac{r_2^2}{r_1^2+r_3^2+r_4^2}+\dfrac{r_3^2}{r_1^2+r_2^2+r_4^2}+\dfrac{r_4^2}{r_1^2+r_2^2+r_3^2}$

(A) 4 (B) 3
(C) 1 (D) 8

SECTION–I(ii) : (Maximum Marks : 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Let $ax^2+bx+a=0$ be the quadratic equation (where $a,b\in R$) has roots α,β. Consider the family of lines $\left(b^2-2a^2+c^2\right)x+8a^2=2\left(b^2-c^2\right)y$ (where c is a real number). If $|\alpha-\beta|=2$ then

(A) The given family of lines always passes through a fixed point $\left(\frac{-4}{7},\frac{2}{7}\right)$

(B) The given family of lines always passes through a fixed point (2 , -4)

(C) Number of lines belongs to the given family which makes a triangle with coordinate axes of area 16 are only 4

(D) Number of lines belongs to the given family which makes a triangle with coordinate axes of area 2 are only 2

6. The vector $\hat{i}+x\hat{j}+3\hat{k}$ is rotated about its initial point through an angle of $\cos^{-1}\left(\frac{11}{14}\right)$ and it's magnitude is doubled. If the vector in the new position is given by $4\hat{i}+(4x-2)\hat{j}+2\hat{k}$ then which of the following can't be the value of x ?

(A) $-\frac{2}{3}$ (B) $\frac{2}{3}$

(C) $-\frac{20}{17}$ (D) 2

7. Let f(x) is a differentiable function such that $u, v \in R$

$2u\,f\left(u+v^2\right)+v^3+u^4v=2v\,f\left(u^2+v\right)+u^3+uv^4$

Consider the curve P defined by

P : $f\left(x+y\right)-x-y+\frac{1}{2}=2f\left(\sqrt{xy}\right)$, then -

(A) Area bounded by curve P with coordinate axis is $\frac{4-\pi}{4}$

(B) Range of the function f(x) is R^+

(C) Number of normals which can be drawn from point (1 ,1) to the curve P will be 4

(D) Length of tangent drawn from origin to the curve P is 1

8. Let $I=\int_0^1\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}\,dx$ and $J=\int_0^1\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$ then which statement(s) are correct

(A) $I+J=2$ (B) $I-J=\pi$

(C) $J=\frac{2+\pi}{2}$ (D) $J=\frac{4-\pi}{2}$

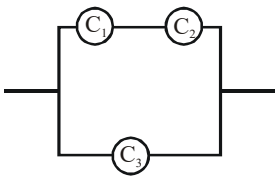
9. If A, B, C are the angles of an acute angled triangle ABC and

$$D=\begin{vmatrix}(\tan B+\tan C)^2 & \tan^2 A & \tan^2 A\\ \tan^2 B & (\tan A+\tan C)^2 & \tan^2 B\\ \tan^2 C & \tan^2 C & (\tan A+\tan B)^2\end{vmatrix},$$

then the value of $\frac{D}{1000}$ can be

(A) –1 (B) 0 (C) 2 (D) 3

10. An electrical apparatus with three components C_1, C_2 and C_3 as in the diagram. The components works independently of each other with probabilities $\frac{8}{10}, \frac{9}{10}$ and $\frac{3}{4}$. For apparatus to be operational current flows from left to right. Let X denote the event that apparatus is operational and X_1, X_2, X_3 denotes respectively the events that the component C_1, C_2 and C_3 are functioning. Which of the following is(are) true ?



(A) P(exactly two components are functioning | X) = 1

(B) If P(X) is $\frac{a}{b}$ (a,b are co-prime) then b – a is 7

(C) $P(X \mid X_3) = 1$

(D) $P(X \mid X_2) = 19/20$

11. Point A lies on x-axis and point B lies on y = x. If PA.PB is minimum where P divides segment AB internally, then (given, P is $\left(1, 2-\sqrt{3}\right)$ and 'O' is origin), -

(A) $OA = 2\sqrt{2} - \sqrt{6} + \sqrt{3} - 1$ (B) $OB = 2\sqrt{2} - \sqrt{6} + \sqrt{3} - 1$

(C) $\angle OAB = 45°$ (D) $\angle OBA = 22.5°$

12. For the function $f(x) = \sqrt{4x^2 - 4x + 2} + |x|$ which of the following holds goods

(A) The least value of the function is $\frac{\sqrt{3}+1}{2\sqrt{2}}$

(B) The least value of function is $\frac{\sqrt{3}+1}{\sqrt{2}}$

(C) The function takes its least value at $x = \frac{3-\sqrt{3}}{6}$

(D) Function has exactly one point of minima

SECTION–II: (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE / THREE DIGIT INTEGER** ranging from 0 to 999, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

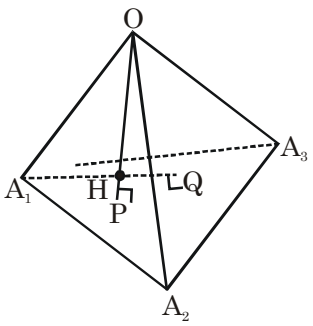
13. Let the function g(x) where $g : R \to R$, be continuous in $[0, \infty)$ and positive in $(0, \infty)$ satisfying the condition $g(1) = 1$ and $\frac{1}{2}\int_0^x g^2(t)\,dt = \frac{1}{x}\left(\int_0^x g(t)\,dt\right)^2$ then the value of $(g(4))^{\sqrt{2}-1}$ is

14. If sum and the difference of the roots of the characteristic equation of a square matrix of order 2×2 are 4 and 2 respectively then the determinant of the matrix will be

15. If $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(2) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}$, $f(x)$ is a quadratic function and its maximum value occurs at a point V. A is a point of intersection of y = f(x) with x-axis and point B is such that chord AB subtends a right angle at V. Let A be the area enclosed by f(x) and chord AB. Find [A] (where [·] denotes the greatest integer function)

16. The value of the series $\dfrac{\sum_{r=0}^{100}(-1)^r\ {}^{100}C_r\,(r)^{50}}{(50)^{100}}$ is equal to

17. Figure shows a tetrahedron $OA_1A_2A_3$. Altitudes OP and A_1Q are drawn from O and A_1 on opposite triangular faces. It is given that OP and A_1Q intersect at H. Let the lengths of OA_1, A_2A_3 be l_1 and l_2 respectively. If shortest distance between OA_1 and A_2A_3 is d and volume of tetrahedron is V. Value of $\dfrac{d\ell_1\ell_2}{V}$ is equal to



18. Let $S_n=\sum_{r=1}^{n} r.2^{\frac{r}{2}(1+(-1)^r)}.3^{\frac{r}{2}(1-(-1)^r)}$. If $S_{20}=\alpha.3^{21}+\beta.2^{22}+\dfrac{391}{288}$, then the value of $\left[\dfrac{32\alpha-9\beta}{4}\right]$ is (where [.] denotes the greatest integer function)

ENTHUSIAST SRG TEST SERIES

TEST - 11 (PAPER 1)	DATE : 12 - 01 - 2022
SYLLABUS	
PHYSICS : FULL SYLLABUS	
CHEMISTRY : FULL SYLLABUS	
MATHS : FULL SYLLABUS	